MALWARE FAMILY CLASSIFICATION

Using a Custom CNN with Attention

TASK 1 REPORT

Image Exploratory Data Analysis and Related Work

Dataset: Malimg Original
Track: Custom CNN + Attention
Course: ICE478
Group: Group02

Submitted by

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1. Introduction

Malware classification is a central cybersecurity problem because malware families continuously evolve through repacking, obfuscation, and code modification. Signature-based detection can be ineffective when new variants do not exactly match known signatures. Image-based malware analysis provides an alternative representation in which the bytes of a malware binary are mapped to grayscale pixel intensities. This transformation allows convolutional neural networks to learn recurring texture and structural patterns without requiring full disassembly.

This project follows the Custom CNN + Attention track. Task 1 explores the Malimg Original dataset, identifies class imbalance and data-quality issues, reviews recent CNN and attention-based malware-classification studies, and defines the research gap that motivates the proposed model. The later tasks will develop representative CNN baselines, a custom attention model, improvement experiments, ablation studies, and explainability analyses.

2. Dataset Description

The Malimg dataset represents malware binaries as grayscale images and organizes them by malware family. The canonical Malimg corpus used in this project contains 9,339 images from 25 malware families. Every image is stored in a class-specific folder, allowing the folder name to be used as the multiclass label.

Property	Description
Dataset name	Malimg Original
Source	Kaggle
Domain	Cybersecurity
Learning task	Multiclass malware-family classification
Standard image count	9,339
Number of classes	25
Image representation	Grayscale malware byte images
Dominant file format	PNG
Repository dataset policy	Dataset files are not committed to GitHub

3. Exploratory Data Analysis Methodology

The EDA notebook was designed to satisfy the required image-analysis items while preserving reproducibility and preventing avoidable memory usage. The main steps are summarized below.

- Recursively enumerate supported image files and infer labels from parent folders.
- Validate every image using PIL and OpenCV and record unreadable files separately.
- Extract width, height, aspect ratio, resolution, file format, image mode, channel count, and file size.
- Measure grayscale brightness, contrast, Laplacian variance, and Shannon entropy.
- Calculate class counts, percentages, the majority-to-minority ratio, and cumulative class concentration.
- Display one labelled sample from every malware family and multiple samples from selected majority and minority families.
- Sample pixel intensities efficiently to avoid loading every image pixel into memory.

- Detect exact duplicate files using SHA-256 hashing and identify any cross-class duplicate groups.
- Export report-ready figures and CSV tables to the Kaggle working directory.

4. Class Distribution and Imbalance

The dataset is highly imbalanced. The largest family, Allaple.A, contains 2,949 images, whereas the smallest family, Skintrim.N, contains only 80 images. The resulting majority-to-minority ratio is approximately 36.86:1. The two Allaple families together account for a substantial proportion of the dataset, which means that overall accuracy can be dominated by large classes.

Measure	Result
Total images	9,339
Total malware families	25
Majority class	Allaple.A (2,949)
Minority class	Skintrim.N (80)
Majority-to-minority ratio	36.86:1
Primary evaluation emphasis	Macro-F1 and class-wise recall

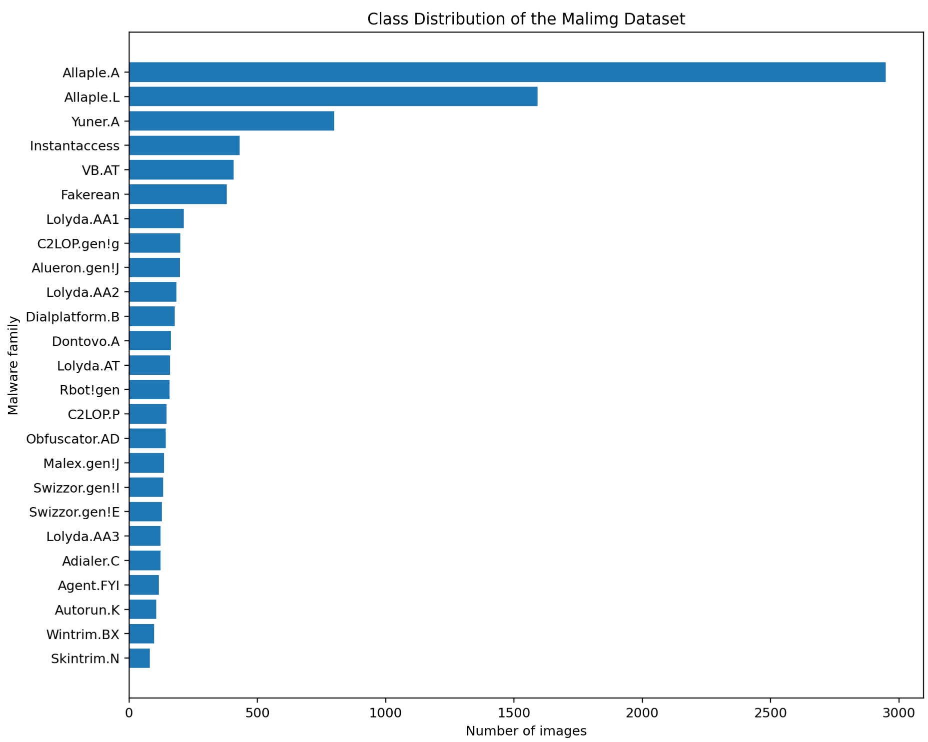


Figure 1. Class distribution of the standard Malimg corpus.

Because of this imbalance, Task 2 should not rely on accuracy alone. The evaluation must include macro precision, macro recall, macro-F1, weighted F1, class-wise metrics, and a confusion matrix. Class weights or another carefully justified imbalance-handling method should be applied only to the training data.

5. Labelled Image Analysis

A labelled grid with one representative image from every family is used to compare broad visual patterns. Malware byte images often contain repeated horizontal bands, dense texture blocks, sparse regions, and family-specific structures. Some families appear visually distinct, while others share similar grayscale textures. These visually similar classes are likely to produce confusion and are important targets for later class-wise error analysis.

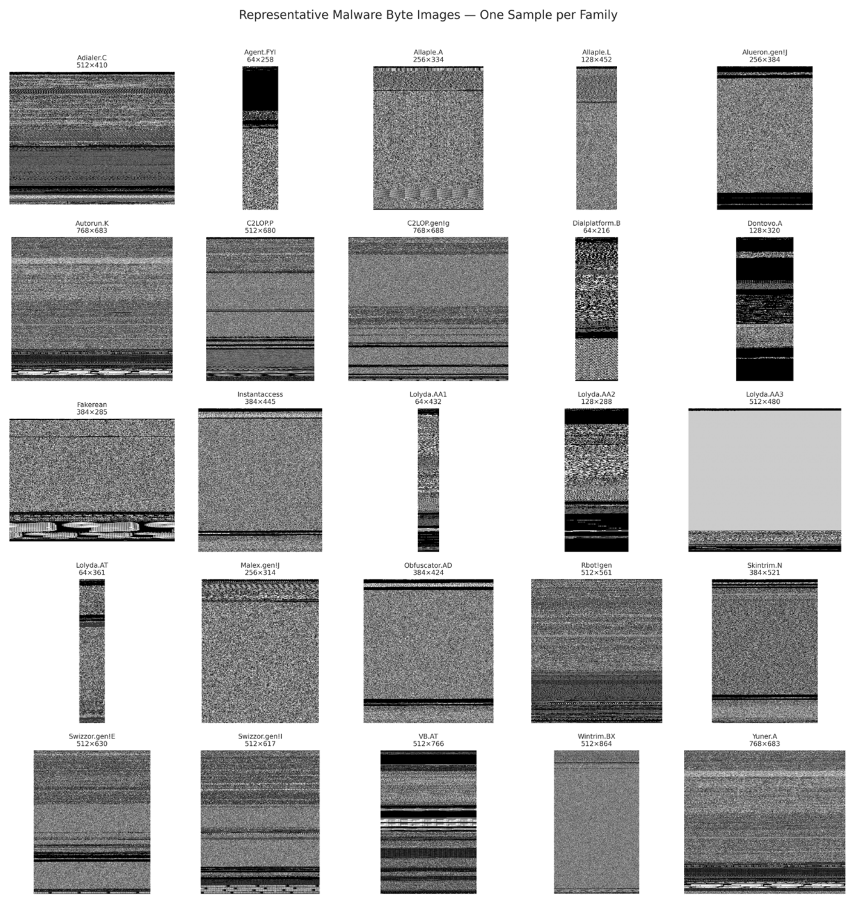


Figure 2. One representative malware byte image from each family.

A second grid should compare several examples from selected large and small classes. This reveals whether a class has consistent internal texture or strong within-class variation. The observation supports the use of attention because the proposed network should emphasize discriminative feature channels and spatial regions rather than treating every location equally.

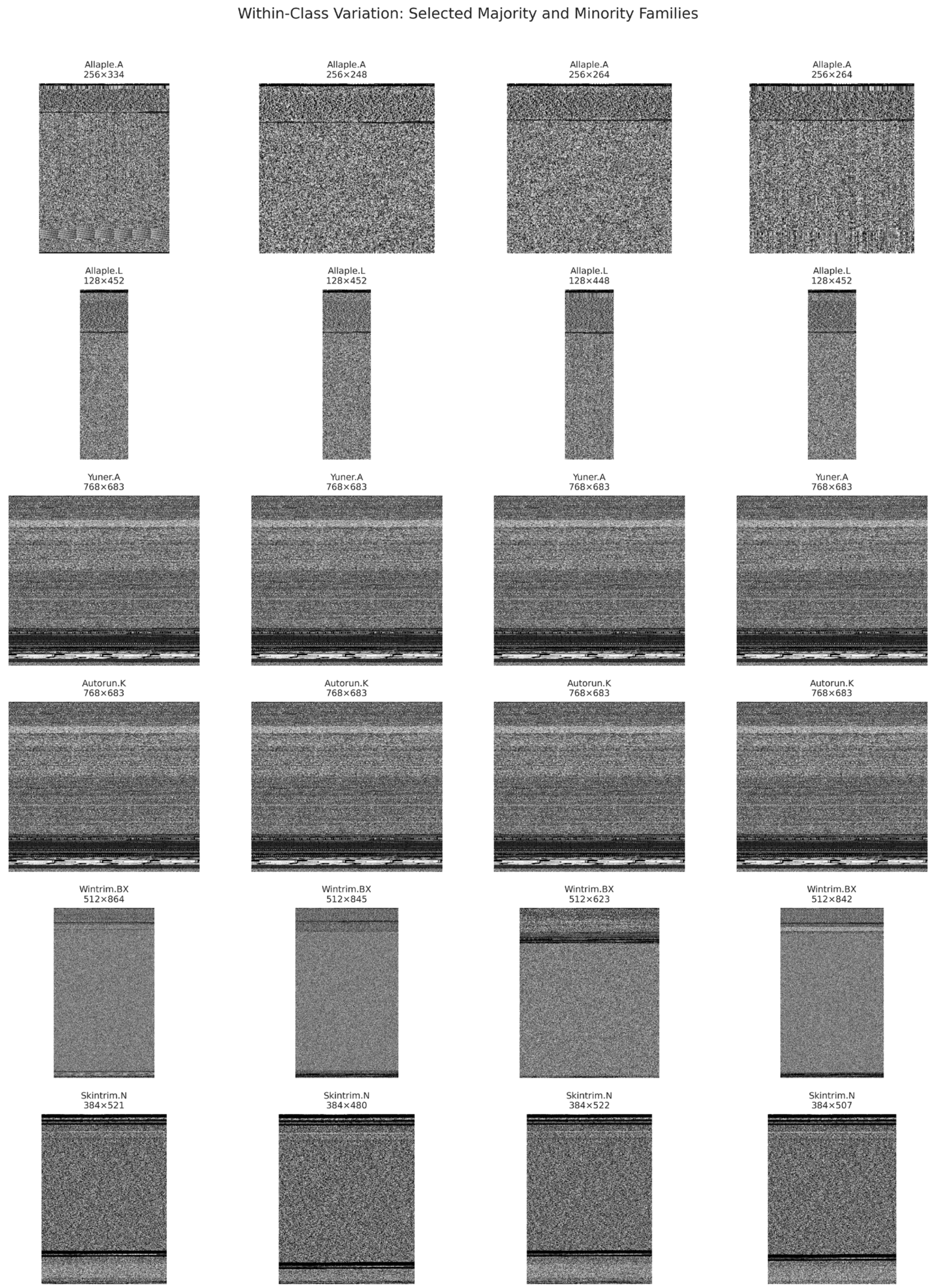
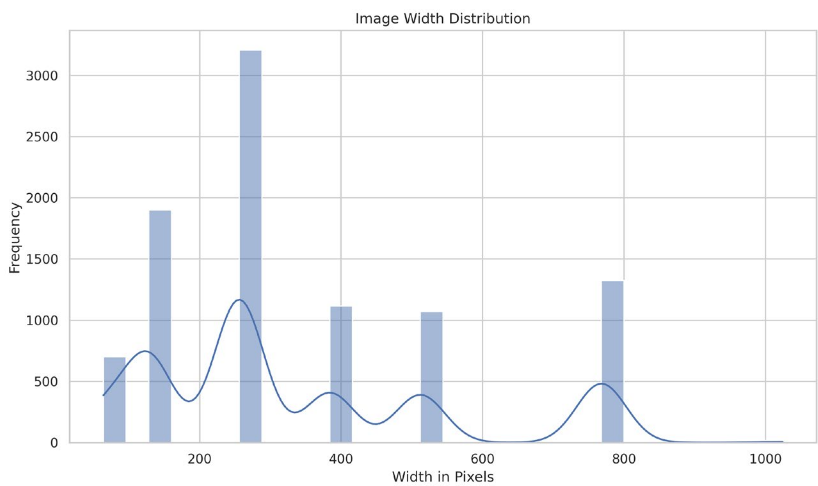


Figure 3. Within-class visual variation for selected majority and minority families.

6. Image Size and Resolution Analysis

Image dimensions are important because CNN models require a consistent tensor shape. The Task 1 notebook measures width, height, aspect ratio, file size, unique resolutions, and IQR-based dimension outliers. The final report values below must be copied from the fully executed Kaggle notebook.

Dimension statistic	Notebook result
Minimum width	64
Maximum width	1024
Mean width	334.17
Median width	256
Minimum height	208
Maximum height	5334
Mean height	436.81
Median height	424
Unique resolutions	835
Dimension outlier count	577



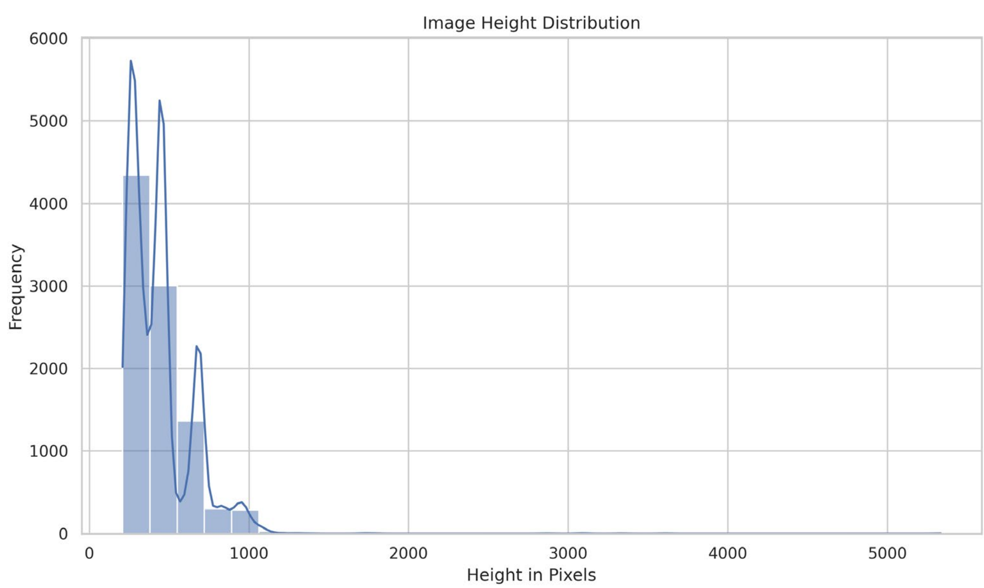


Figure 4. Width and height distributions of Malimg images.

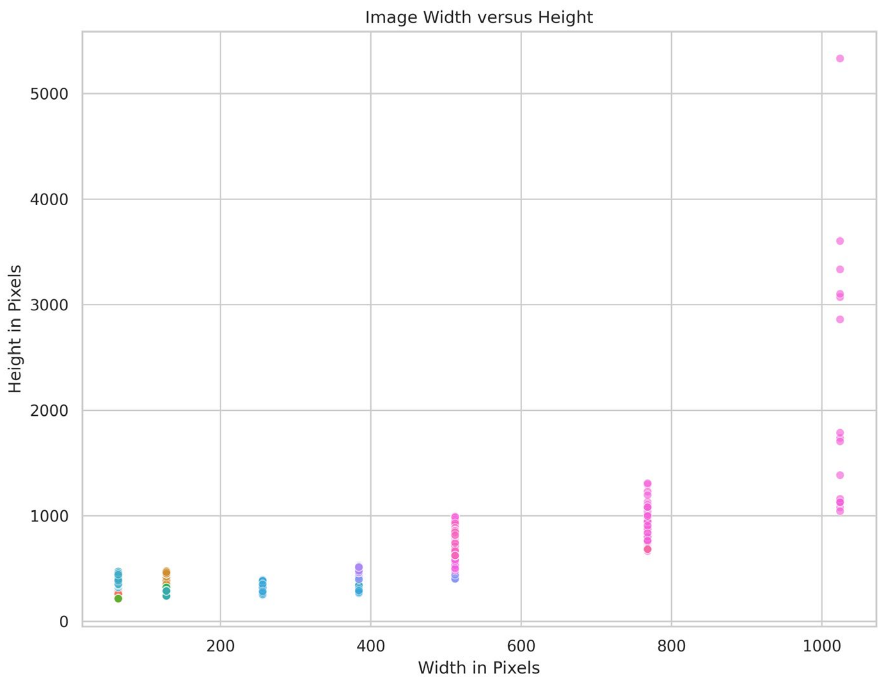


Figure 5. Relationship between image width, height, and aspect ratio.

If multiple resolutions are present, every model must use the same resizing policy to ensure a fair comparison. Resizing should be applied after data splitting and consistently across training, validation, and test images. The chosen input size will be reported explicitly in Task 2.

7. Pixel, Channel, and Image-Quality Analysis

Malimg images are grayscale byte representations, so the intensity of each pixel corresponds to a byte value rather than natural-image colour. The pixel histogram reveals the global distribution of byte intensities. Brightness and contrast summarize image intensity characteristics, Laplacian variance provides a texture-sharpness indicator, and Shannon entropy measures pixel-level complexity.

Quality measure	Notebook result
Dominant channel count	1 (grayscale)
Mean brightness	113.57
Median brightness	121.5
Mean contrast	79.6
Median contrast	77.2
Mean Laplacian variance	103848.93
Mean Shannon entropy	7.226
Dominant image format	PNG

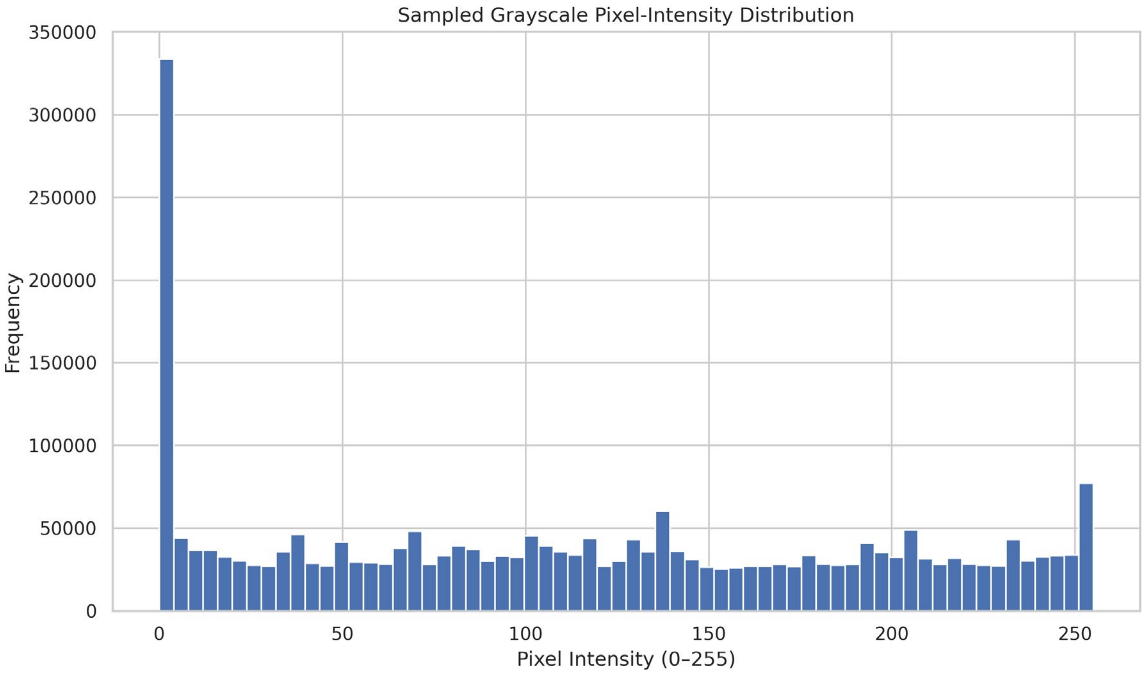


Figure 6. Sampled grayscale pixel-intensity distribution.

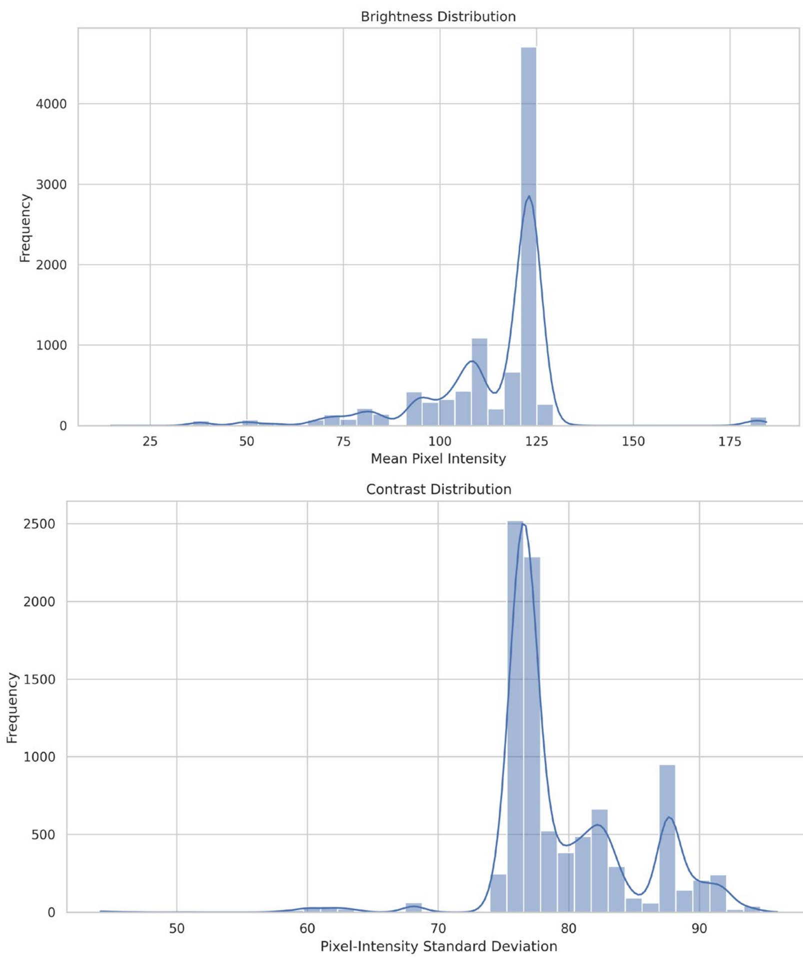


Figure 7. Distribution of a selected image-quality measure.

These measures are descriptive rather than direct indicators of malware severity. In particular, the Laplacian score should not be interpreted exactly as photographic blur because malware images are synthetic byte visualizations. The values are useful for identifying unusual textures and comparing families, but they do not carry ordinary visual semantics.

8. Data-Quality and Leakage Analysis

The notebook checks missing metadata, unreadable images, duplicate dataframe rows, duplicate paths, and exact duplicate image content. SHA-256 is used to identify byte-for-byte identical files. Exact duplicates are particularly important because the same image must not appear in both training and evaluation subsets.

Data-quality check	Notebook result
Corrupt or unreadable images	0
Missing metadata values	0
Duplicate dataframe rows	0
Duplicate image paths	0
Exact duplicate groups	260
Duplicate images beyond first copy	895
Cross-class exact duplicate groups	0

Corrupt images, if any, should be excluded and documented. Cross-class exact duplicates require manual inspection because identical image content assigned to different labels can indicate labeling inconsistency. All cleaning decisions must be reproducible and reflected in the final metadata file.

9. Related Work

Six recent studies were reviewed because they employ CNNs, channel or spatial attention, cross-modal attention, or multi-head self-attention for malware-image classification. Their reported results cannot be ranked directly because the studies use different inputs, splits, classifiers, imbalance treatments, and validation protocols. The complete comparison is provided separately in Group02_Malimg_related_work_table.pdf.

Study	Architecture / Method	Attention Type	Reported Result
Van Dao et al. (2022)	Lightweight CNN + attention-VAE features + Random Forest	CBAM-style channel + spatial	99.40% accuracy on unbalanced Malimg
Kim et al. (2023)	Cross-modal CNN using images + structural entropy	Cross-modal attention	99.09% accuracy; F1 = 0.976
Ben Abdel Ouahab et al. (2023)	Vision Transformer compared with CNN	Multi-head self-attention	98.38% ViT accuracy
Dong (2024)	RGB conversion + GAN oversampling + simplified ViT	Multi-head self-attention	96.97% accuracy; F1 = 97.02%
Ayturan (2026)	Pretrained ResNet-50 + CBAM	Channel + spatial attention	99.36% Malimg test accuracy
Uddin et al. (2026)	Parallel CNN + ECA + LSTM	Efficient Channel Attention	99.23% accuracy

The studies show that attention can support strong malware classification; however, attention is usually introduced together with other major architectural or preprocessing changes. This makes it difficult to determine whether the gain comes specifically from attention.

10. Research Gap

Existing malware-image studies demonstrate strong performance using CNNs, CBAM, channel attention, cross-modal attention, LSTM, and Vision Transformers. However, many approaches combine attention with a variational autoencoder, external classifier, multimodal structural features, GAN-generated samples, a large pretrained backbone, or an LSTM. Consequently, the independent contribution of attention is often confounded with other design changes.

Limited work provides a controlled comparison of the same custom CNN without attention, with channel attention, with spatial attention, and with combined channel-spatial attention. In addition, class-wise performance under severe imbalance, five-fold cross-validation, significance testing, systematic attention ablation, Grad-CAM, and LIME are not consistently reported. This project addresses that gap through a matched and leakage-free experimental design.

11. Task 1 Conclusion

Task 1 established the principal characteristics of the Malimg dataset. The corpus contains 9,339 images from 25 malware families and exhibits severe imbalance, with Allaple.A containing 2,949 images and Skintrim.N containing only 80. The EDA pipeline also measures image dimensions, aspect ratios, pixel intensities, image-quality characteristics, corrupt files, and exact duplicates.

These findings indicate that Task 2 must use a leakage-free stratified split, consistent resizing and normalization, appropriate imbalance handling, and class-wise evaluation. The related-work analysis further motivates a controlled custom CNN + attention experiment in which the contribution of channel and spatial attention can be isolated from other changes. The final highlighted numerical values in this document must be updated from the fully executed Kaggle notebook before PDF export and GitHub submission.

References

[1] T. Van Dao, H. Sato, and M. Kubo, “An Attention Mechanism for Combination of CNN and VAE for Image-Based Malware Classification,” IEEE Access, vol. 10, pp. 85127-85136, 2022. doi: 10.1109/ACCESS.2022.3198072.

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[7] Malimg Original, Kaggle dataset. Available: https://www.kaggle.com/datasets/ikrambenabd/malimg-original.